

Supplement to Separate Ways, V3

This document holds calculations the main argument does not need. Each section states its own assumptions, and several of them assume more than the paper does. Nothing here is a result of the Lorentzian two-copy construction: printing a calculation beside that construction does not make it one.

V2's programme-comparison section is not part of this submission. It stays in the working record.

One note on notation, following the papers these sections examine: A is the antipodal map, which the main paper writes α everywhere except its Appendix C.

S1. Bilocal noise and the non-traversable bridge reading

In the free two-patch sector the partner correlation has a thermofield interpretation. It does not come with a bridge. The fold adds no interaction that would make the two exteriors traversable, and equality with the ordinary restricted state uses the local premises of main Appendix A.1. On the central worldline the conformal-scalar image kernel equals the half-KMS-period cross-leg kernel; away from that worldline you have to keep the transverse antipode as well. Nothing about interacting horizons or evaporation follows.

A normalised bilocal scalar-probe toy makes the noise split explicit. Let

$$F_{\text{even}} = \frac{F_1 + F_2}{\sqrt{2}}, \quad F_{\text{odd}} = \frac{F_1 - F_2}{\sqrt{2}}.$$

On the central comoving worldline in the conformal Bunch-Davies state, with the image frame carried by the antipode,

$$N_I(t) = -N_0(t - i\beta/2), \quad \beta = \frac{2\pi}{H}.$$

The integrated coefficients are $A_0 = H^3/(12\pi^2)$ and $A_I = -H^3/(12\pi^2)$, so the normalised even port has zero integrated noise and the odd port has $H^3/(6\pi^2)$. Both coefficients and both port eigenvalues are computed in, and the off-axis case in. Their spectra are

$$S_{\text{even}} = \frac{\omega(\omega^2 + H^2)}{12\pi} \tanh\left(\frac{\pi\omega}{2H}\right), \quad S_{\text{odd}} = \frac{\omega(\omega^2 + H^2)}{12\pi} \coth\left(\frac{\pi\omega}{2H}\right),$$

Equivalently, $S_{\text{even}} = \rho(1 - 2n_F)$ and $S_{\text{odd}} = \rho(1 + 2n_B)$ at temperature H/π . At $\omega = H$ and $H/2$, the even port lies 8.285% and 34.421% below the vacuum, while the odd port lies 9.033% and 52.487% above it. The main text's average/difference variables have a different normalisation: $F_c = F_{\text{even}}/\sqrt{2}$ and $F_q = \sqrt{2}F_{\text{odd}}$, so their spectra gain factors $1/2$ and 2 .

This is a reading of a postulated bilocal probe, not a prediction. The action contains no worldline-to-image coupling, and the central-worldline KMS identity does not extend to an assertion that the full antipode is only a time shift.

S2. The RP3 topology comparison

This comparison claims nothing about the existence of the branches or about their decoherence, both of which are standard quantum cosmology. Identifying the contracting branch

with the CPT image of the expanding one is a further assumption, strictly stronger than reality of Ψ , and it is named as one. What the comparison does supply is a number. The RP^3 quotient deletes the even- n harmonics, and $n = 2$ alone carries 76% of the exponent, so the restriction raises A_{dec} from 1.954 to 2.595, a change of 32.84%, and the proper time Ht from 1.290 to 1.608, a change of 24.63%. All four amplitudes and both percentages come from. The folded saddle is tree-level degenerate with the no-boundary cap, and the difference is one loop.

Falsifiable content of the clock comparison: none. Exactly one harmonic, $n = 2$ on the cover and $n = 3$ on RP^3 , leaves the horizon before the tick, and even at $|\Omega_K| = 0.005$ its wavelength is seven times the diameter of the observable universe. Every pre-tick mode is super-horizon, so there is nothing here a measurement could catch. Adopting the RP^3 spatial quotient on its own would instead face the standard cosmic-topology search, which only bites for $\Omega_{tot} > 1.166$. The Lorentzian cover this paper defends does not acquire that topology from the antilinear field relation alone.

Which fields contribute is worth stating, because most of the Standard Model does not. In the free, unbroken, massless approximation the 90 fermionic and 24 transverse gauge degrees of freedom are conformal and contribute nothing to this branch-overlap exponent. The four real Higgs components depend on a curvature coupling that has to be chosen separately: massless conformal coupling gives zero, minimal coupling gives four scalar contributions. Masses and interactions break that idealisation and are not computed here. The tensor tower runs from $n \geq 3$ with multiplicity $2(n^2 - 4)$, so the graviton is not two scalar towers.

The mass dependence is exact. $A_{dec}(m/H)$ takes the values 1.954, 1.960, 2.011, 2.997, ∞ and 3.291 at $m/H = 0, 0.1, 0.3, 1, \sqrt{2}$ and 3. The infinity is not a divergence but the absence of an effect: a minimally coupled scalar of mass $m^2 = 2H^2$ is the conformal scalar, and it makes no arrow at all. Below $m = \sqrt{2}H$ the mass-dependent coefficient is $(\pi/48)(m^2/H^2 - 2)^2$, which is why it vanishes there.

At large A the massless-field decoherence law approaches $-\log|D| \simeq (N_{eff}/18)\mathcal{N}_H$, with $\mathcal{N}_H = (3\pi/2)(aH)^3$ the number of Hubble volumes in a closed S^3 . That asymptote is not exact near the throat or at the first e^{-1} crossing, which is where A_{dec} sits.

S3. Euclidean boundary conditions and the CP cost

This comparison imposes additional CP-sector and Euclidean boundary conditions. Neither follows from the defended Lorentzian cover.

A bang mirror forbids the $\theta G\tilde{G}$ term, which solves strong CP without an axion. That is Boyle, Teuscher and Turok's result [1], and the other leg costs the same. Put the Standard Model on the unorientable Euclidean section with a Pin^+ structure and the single reflection has to be a symmetry of the Lagrangian rather than of the spectrum alone, so CP violation must be spontaneous. García-Etxebarria and Montero put it plainly: "This theory would then make sense in unorientable spacetimes, as long as these admit fermions" [2], their §3.4, which also establishes $\Omega_5^{\text{Pin}^+}(BG_{\text{SM}}) = 0$. Reading CP as a discrete gauge symmetry on a nonorientable spacetime is Choi, Kaplan and Nelson's [3]. The sector this needs is a generic one: vector-like quarks from 1.2 TeV to tens of TeV, with $\bar{\theta}$ near the present neutron-EDM limit in the maximal corners [4]. So n2EDM and nEDM@SNS test it.

The two legs then differ in a way a measurement could catch, and that part is new. On the bang mirror a CT-odd CP scalar obeys $\varphi(-t) = -\varphi(t)$, so $\varphi(0) = 0$: the wall is the bang, and there is nothing in our sky. On the elliptic leg $\varphi(Ax) = -\varphi(x)$ makes φ odd on the late-time S^3 , and Borsuk-Ulam then forces an antipodally symmetric zero set separating every point from

its antipode. So the elliptic and CT-odd conditions together force one cosmological-scale CP wall, even if CP breaking precedes inflation.

That wall has a price. It is harmless only for $\Lambda_{\text{CP}} < \Lambda_* = (\rho_c R_H)^{1/3} = 29.5 \text{ MeV}$, computed in, which recovers the Zel'dovich-Kobzarev-Okun bound [5] (§S11.10). Above that threshold the wall has to sit $(\Lambda_{\text{CP}}/\Lambda_*)^3$ Hubble radii beyond the horizon, which inflation must supply as

$$\Delta N = 3 \ln \left(\frac{\Lambda_{\text{CP}}}{29.5 \text{ MeV}} \right) \quad \text{extra e-folds,}$$

giving 10.6 extra e-folds at 1 GeV, 31.3 at 10^3 GeV and 65.8 at 10^8 GeV .

An elliptic fold with a Nelson-Barr scalar at 10^8 GeV therefore needs about 126 e-folds of inflation: the usual sixty, plus those 65.8. That is a bound on inflation, not on CP physics. The CT-odd assumption, the generic wall position and the neglect of the wall's repulsive gravity are all stated above. The alternative is not free either, since gauged and spontaneously broken CP has a domain-wall problem [6], though no global anomaly [7].

S4. CPT-regular modes and complexity

The bang of the two-sheeted radiation-plus- Λ solution [8] is an isotropic singularity in Goode and Wainwright's sense [9–14], which the authors themselves note [15]. What this section adds is a positive-definite invariant.

Build it from the electric and magnetic parts of the Weyl tensor, in the manner of Senovilla [16] and of Clifton, Ellis and Tavakol [17], and the complexity $\square$ vanishes at the bang like η^2 for the CPT-regular modes and diverges like $|\eta|^{-1}$ for the irregular ones. **The bang is a complexity minimum with the CPT condition and a maximum without it** (Figure 2; fitted exponents +1.9999989 and +1.9999975 against -0.999994 and -1.000000 ; §S11.9 and the companion calculations).

Two legs of Barbour, Koslowski and Mercati's Janus-point structure [18–20], still Newtonian in 2026 [21], transfer exactly at linear order. The third, $D = 0$, fails, so the full identification stays an analogy. We found no cross-citation between the two literatures.

S5. A discrete-curvature mechanism under test

Whose claim this is. What follows tests a proposal of Deng and Handley's, not a prediction of this framework. Nothing here requires quantised curvature. The elliptic reading halves their allowed set and the data then close what is left, and that is recorded because closing a line is worth more than leaving it open. A reader who wants only this paper's own commitments can use the commitments table in main §3.6.

Deng and Handley impose CPT at the future conformal boundary as well as at the bang, which restricts the allowed modes; in a closed universe the wavevectors are already integers, and matching the two ladders quantises the curvature [22], building on Lasenby, Handley, Bartlett and Negreanu [23]. Their 2025 refit with matter and a Boltzmann hierarchy against a filtered Planck chain [24] gives $\Omega_K \simeq -0.61/N^2$ for $N = 3 \dots 7$ as an asymptotic rule, accumulating at flatness; it matches their table from $N = 5$ up and misses the $N = 3$ entry by 12%; we reproduce it to 0–4%, and obtain a closed form for their 2024 matching integral, checked against quadrature to 2.4×10^{-16} (§S11.7).

The elliptic reading halves the set. Degree- ℓ scalar harmonics on S^3 carry antipodal eigenvalue $(-1)^\ell$, so RP^3 keeps $\nu = \ell+1$ odd, while the periodic ladder runs over $N, 2N, 3N, \dots$; if N were even the intersection would be empty, leaving no perturbations at all. Hence N is odd, deleting $N = 4$, the member their own χ^2 prefers. (This N is Deng and Handley’s mode-ladder integer and is unrelated to the equation-of-state ladder’s N in §S11.8, which is also required odd, for a different reason.)

The confrontation. Deng and Handley compare to Planck alone and defer a joint analysis, so this comparison is ours. Against DESI DR2 BAO plus CMB, $\Omega_K = +0.0023 \pm 0.0011$ [25]; against Planck 2018 alone, -0.0106 ± 0.0065 [26]; our own refit of the DESI DR2 points with compressed Planck priors reproduces the first, $+0.00275 \pm 0.00106$ (2.55σ).

N	Ω_K	vs Planck alone	vs DESI DR2 + CMB	in RP^3 ?
3	-0.076	10.1 σ	71.2 σ	kept
4	-0.039	4.4 σ	37.5σ	deleted
5	-0.024	2.1 σ	23.9 σ	kept
6	-0.016	0.8 σ	16.6 σ	deleted
7	-0.012	0.22σ	13.0σ	kept
∞	0 $^-$	1.6 σ	2.09 σ	limit, never attained

Figure 3 plots the set against both datasets.

Their preferred $N = 4$ model, run against DESI DR2 with every calibration input profiled out so that only the shape of the distance-redshift relation is tested, gives $\Delta\chi^2 = +221$ on twelve shape degrees of freedom. Its $H_0 = 55.0$ sits 15.6σ from SH0ES’s $73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [27], against 5.2σ for flat ΛCDM .

And no reading can produce a positive Ω_K , because $\Omega_K = -3\tilde{\kappa}\sqrt{\Omega_\Lambda\Omega_r}$ with $\tilde{\kappa} > 0$ by construction. The integer condition exists only for compact spherical sections, and RP^3 is one of those, so the elliptic reading does not escape it either. Even exact flatness is 2.09σ from the data.

Two caveats against us. The Planck-alone column summarises a strongly non-Gaussian posterior, so Planck alone does genuinely prefer -0.039 over flatness. And the sparse branch might be thought inadmissible for deleting most primordial modes, though their 2025 paper answers that. The cheapest decisive test is to re-run their filtered-chain χ^2 at mode spacing $2N$ with N odd inside the existing quantised-power-spectrum pipeline [28–29]. That needs their modified CLASS and is not done here.

S6. A primordial-tilt mechanism under test

Whose claim this is. The tilt formula below is Turok and Boyle’s and contains nothing from the fold. The tension the table reports is with their mechanism. The separately adopted BFT production and mass-sector commitments do not go through it.

Turok and Boyle derive a primordial spectrum from dimension-zero scalar fields and obtain a tilt with no free parameters, subject to two stated theoretical assumptions [30]. We reproduce the derivation exactly. Their eq. 13 prefactor $3^25^3/(7(2\pi)^4) = 0.103118125$ agrees to eight decimals, and the tilt is

$$n_s = 1 - 7\alpha_3(M_P)/\pi = 0.957888.$$

Four things fix that number: the QCD coefficient $11 - 2n_f/3 = 7$; $\alpha_3(M_P) = 0.0189$ [31]; the assumption $p = 2$, which the authors flag; and constancy of n_s across 63.5 e-folds. The last

of those carries weight, because a running α_3 reaches its Landau pole 47.5 e-folds below M_P . recomputes the tilt from $\alpha_3(M_P)$ and every tension in the table below from the quoted n_s and σ .

dataset	n_s	$(n_s - 0.957888)/\sigma$
Planck	2018 0.9649 ± 0.0042	1.67σ
TT,TE,EE+lowE+lensing		
ACT DR6 alone	0.9666 ± 0.0077	1.13σ
SPT-3G D1 alone	0.9510 ± 0.0110	-0.63σ
P-ACT	0.9709 ± 0.0038	3.42σ
CMB-SPA	0.9679 ± 0.0033	3.03σ
P-ACT-LB	0.9743 ± 0.0034	4.83σ
P-ACT-LB2	0.9752 ± 0.0030	5.77σ
CMB-SPA + DESI DR2	0.9726 ± 0.0028	5.25σ

Figure 4 plots the displacements.

[26,32–34]. This is posterior tension, not formal exclusion: these are displacements between a fixed prediction and marginal posteriors in base Λ CDM, not a likelihood ratio for a nested model. This specified comparison uses base Λ CDM; the fold construction alone does not derive universal values of N_{eff} or the running. Freeing N_{eff} removes the tension entirely, but the value needed is $\Delta N_{\text{eff}} \approx -0.35$ to -0.82 , less radiation than the Standard Model, which the dimension-zero scalars do not supply [35]. The mechanism is also over-constrained internally: requiring the measured amplitude at $N_0 = 36$ with Standard-Model couplings forces $n_s = 0.9498$, 7.2σ on the other side of P-ACT-LB. As a two-observable predictor it does not survive; as a one-observable fit it survives and predicts nothing.

Mechanism, not structure. Nothing in the CPT-symmetric structure, and nothing in the fold, enters the tilt formula: it is built from the high-temperature trace anomaly, the QCD β -function and an assumption about which factor of α carries the scale dependence, as the authors say in their own conclusion. The tension concerns this tilt mechanism. It does not change the separately adopted BFT production calculation, its primordial-tensor commitment or the massless-neutrino consequence of exact stabilisation. The 4.8×10^8 GeV abundance-calibrated mass is likewise unaffected. We record that the dimension-zero scalars are under independent theoretical dispute [36], unanswered at the time of writing, and we found no second live test. Early-galaxy abundances require an explicit initial-condition, growth and astrophysical model before they can test the specified cosmological implementation [37–40].

S7. A future-horizon low-quadrupole conjecture

The tensor response of §S9 has no scale at the bang. For the fixed Planck-like expansion history and observer epoch used here, the remaining conformal distance from today is $\eta_f = \int_1^\infty da/(a^2 H) = 1.14965 c/H_0 = 5.114$ Gpc. Its inverse is $k_c = 1/\eta_f = 1.96 \times 10^{-4}$ Mpc $^{-1}$, near the quadrupole scale. As a conditional scalar-spectrum shape test, choose $S(k) = (k\eta_f)^4/[1 + (k\eta_f)^4]$; neither this scalar multiplier nor its future-boundary origin is derived. A simplified Sachs–Wolfe plus late integrated-Sachs–Wolfe calculation gives TT ratios 0.734, 0.862, 0.936 and 0.969 at $\square = 2$ –5, and 0.998 at $\square = 10$. The EE response has not been calculated. Applied to the published Planck 2018 spectrum estimates [26], an idealised independent full-sky cosmic-variance proxy gives $\Delta(-2 \ln L) = -1.030$ and quadrupole lower-tail probabilities 0.0469 and 0.0884. These are not an evaluation of the official Planck likelihood. The chosen shape would not account for the whole deficit, and it leaves the $\square \approx 20$ –30 dip untouched.

The radiation- Λ Baker-Akhiezer mode with its specified global sheet choice fails a global remaining-distance-only law. The follow-up script propagates its fold data and samples its absolute power; it does not construct the independent cap-local reference proposed in the earlier calculation. Its reported 78%, 62% and 24% spreads used $\pm 12\%$ bins around nominal q . Comparisons matched exactly at $q = 0.3, 0.6, 1.0$ also fail to collapse, with spreads 66.6%, 64.8% and 23.0% on the six- κ audit grid. Here $q = \kappa(T - x)$ measures conformal event-horizon distance; $q \simeq k/(aH)$ only near the cap. The late normalised tensor power is $F = \kappa^2/\sqrt{1 + \kappa^4}$, while the chosen scalar multiplier uses its square. A physical scalar or final-state mechanism and the cap-local-reference calculation remain unestablished. In a universe with matter, the corresponding transition is the matter- Λ crossover, whose conformal scale is of order the horizon today; the crease law has not been recomputed on that background here.

S8. Restricted exact- Λ analysis

This calculation supports the exact- Λ baseline only within a stated metric-only action class. Let K be the expansion scalar and $s = \log(K_c/K)$. The added action contribution is

$$S_{\text{add}} = \int dt d^3x N \sqrt{h} f(K), \quad f(K) = -\rho_c W(s).$$

Its homogeneous energy density is $\rho_c(W + W_s)$, not $\rho_c W$. For arbitrary spatial lapse smearings, the exact shear-retaining Hamiltonian bracket contains

$$\{H[N], H[M]\} = H_i[\dots] - \int d^3x \sqrt{h} \rho_c \frac{W_s + W_{ss}}{K^2} (ND^i M - MD^i N) D_i K.$$

Requiring off-shell closure for this full class of refoliations gives $W_s + W_{ss} = 0$, hence

$$W(s) = W_0 + C e^{-s}.$$

Because $e^{-s} = K/K_c$, the second term contributes to the action in proportion to $\int dt d^3x N \sqrt{h} K$ and is a time-boundary term, up to spatial boundary terms. The surviving bulk term is constant. For a homogeneous metric-only Lagrangian $a^3 f(\theta)$, with $\theta = 3H$,

$$\rho = \theta f'(\theta) - f(\theta), \quad \rho + p = -\dot{\theta} f''(\theta).$$

The constant term therefore has $p_\Lambda = -\rho_\Lambda$ even in a matter-filled decelerating universe. With separate conservation, $Q = 0$ and $\rho_m \propto a^{-3}$.

The scope matters. Spatially constant lapse smearings or a constant-mean-curvature slice make the displayed anomalous term vanish for any W . Whether a reduced TDiff or unimodular completion requires the full lapse class is a separate constraint problem. The result above is therefore a restricted off-shell closure result, not a theorem about every covariant dark-energy theory.

S9. Tensor crease calculation

On the exactly solvable radiation- Λ background, $y_x^2 = 1 + y^4$ in conformal time x (units of the radiation- Λ crossover time η_*), the tensor mode equation is the $n = 1$ Lamé system $u'' + (\kappa^2 - 2y^2)u = 0$, $\kappa = k\eta_*$, and the solution selected by its future frozen branch has fold impedance $\Omega^2 = z + z^{-1}$, $z = \kappa^2$, and late normalised tensor power $F = z/\sqrt{1 + z^2}$,

reproduced by direct Wronskian-normalised integration. The selecting condition is a Baker-Akhiezer one, imposed at the cap rather than at the crease.

That power is one logistic variable read several ways. With $s = \log z$ it is $F^2 = \sigma(2s)$, the associated response crossover is $\tanh s$, and the Fisher information is $\text{sech}^2 s$. On the lemniscatic curve $j = 1728$, which is what this background is, that Fisher information is a regular degree-eight Belyi map, $64/u^4$ in the quotient coordinate $u = p/z$, verified independently.

The result is exact, and it is also unobservable, which is worth saying in the same breath. At the observed Λ the amplitude it multiplies is $2H_\Lambda^2/(\pi^2 M_{\text{Pl}}^2) \simeq 5 \times 10^{-122}$, and the crossover $k_c = \eta_*^{-1} = 2 \times 10^{-5} \text{ Mpc}^{-1}$ lies beyond the horizon. Nothing observable follows from it at the bang crease. The tensor shape follows from the stated background and the future-branch condition; the action does not select that condition, and no observed-scale realisation is derived here.

S10. Detailed information-criterion and laboratory comparisons

The following accounting is supplementary. Information criteria organise tradeoffs but do not decide physical truth.

Give a model enough adjustable numbers and it will fit anything. A curve with ten free coefficients can be made to pass through any ten points, exactly, every time; and the fact that it does tells you nothing about the world. So agreement with data is never impressive by itself. What is impressive is agreement bought cheaply, with few adjustments. Information criteria put a price on each adjustment and ask whether the improvement in fit was worth what it cost.

There are two prices in common use and they answer different questions. Akaike's criterion asks whether the model will predict the next measurement well, and charges 2 for each free parameter. Schwarz's, usually written BIC, asks whether a parameter is real or is merely soaking up noise, and charges $\ln n$ for each one, where n is the number of data points; with about 1600 points that is a rent of roughly 7.4 per parameter rather than 2. In both, a negative number favours the model being tested and a positive number favours the simpler alternative it is being tested against.

Against ΛCDM with a free neutrino mass sum, the adopted exact-stabilisation BFT sector fixes the sum from oscillation inputs once normal ordering is chosen. The absolute-mass likelihood below does not fit that value. This comparison concerns the restricted cosmological submodel; it does not isolate observational evidence for the geometric fold.

A parameter accounting, since "minimal" is a claim about parameters. Against ΛCDM with Σm_ν free, that sector fixes Σm_ν at 58.8 meV and pays $\Delta\chi^2 = +3.0$ on the geometry-only likelihood summarised in main §3.3 (1580 Pantheon+ supernovae with their full covariance, 13 DESI DR2 BAO points, two compressed Planck priors), while $w_0 w_a \text{CDM}$ gains $\Delta\chi^2 = -9.1$ for two more parameters. With the Akaike penalty of 2 per parameter this is $\Delta\text{AIC} = +1.0$ for the mass-restricted submodel against ΛCDM and -5.1 for $w_0 w_a \text{CDM}$: a tie and a loss. With the Schwarz penalty $\ln n = 7.37$ per parameter, the criterion that asks whether a parameter is real, not whether it helps prediction, the order reverses: $\Delta\text{BIC} = -4.4$ for the mass-restricted submodel against ΛCDM and $+5.6$ for $w_0 w_a \text{CDM}$, so on this likelihood the submodel is preferred over both. Two caveats travel with the second number. The supernova points are correlated, so $\ln n$ overstates the penalty by a factor the effective sample size would reduce, moving the result toward the AIC end; and the CMB-inclusive likelihoods are harder on both sides. With DESI+CMB, fixing Σm_ν at the oscillation floor costs $\Delta\chi^2 = 7.2$ [41], so the submodel's ΔAIC against ΛCDM is $+5.2$ and BIC prefers it only if the effective number of independent data points exceeds 1300; and $w_0 w_a \text{CDM}$ gains $\Delta\chi^2 = -10.7$ to

−21.0 depending on the supernova sample [25], which BIC rewards only if that number is below 210 (Pantheon+) to 36,000 (DESY5). Since a CMB-inclusive likelihood has no simple count of independent points, those break-even values are reported rather than a verdict. The reference varies neither r nor w , so the adopted cosmology’s $r = 0$ and Λ earn nothing here; against models that vary them it earns more. The Turok-Boyle tilt mechanism, which would fix n_s as well, costs $\Delta\chi^2 = 23$ against the P-ACT-LB tilt of §S6 and loses under either penalty; the structure does not carry it.

S11. Supporting calculations for supplementary mechanisms

These calculations support optional comparisons. None of them is a premise of the main argument.

S11.1 Where a crease can sit, and what each position costs. The classification is placed here because main §2.1 does the choosing. Which fold one takes is fixed by the set of points the involution leaves alone, and that set is the crease. The table below is the classification.

position	involution	fixed set	what it costs		verdict
no crease	elliptic A on dS	empty (free)	nothing under premises	locally, stated	conditionally silent (main Appendix A.2)
codim 1, spacelike	$t \rightarrow -t$	the whole spatial slice	a mirror boundary condition; no tension		the best-studied fold
codim 2	modular J	the bifurcation 2-sphere	the quotient-respecting state is Rindler at $2T_U$		dead as a smooth one-copy fold
codim 1, timelike	reflection in a bulk coordinate	a brane worldvolume	negative tension, and a ghost		not a healthy fold
two copies	antilinear pairing on the cover; $\alpha = J \circ P_\perp$ in the stated de Sitter construction	none	standard closed-time-path algebra with a specified global twist		compatible with the demonstrated physical free-graviton sector in main §2

S11.2 The two involutions. Sewell’s theorem gives the modular reflection of a wedge as $(t, w; \mathbf{y}) \rightarrow (-t, -w; \mathbf{y})$ [42]; Borchers and Buchholz specialise it to de Sitter, where the reflection reverses X_0 and X_1 and fixes the transverse coordinates [43]. The antipode reverses all five. Hence $A = J \circ P_\perp$ with $P_\perp$ acting on harmonics as $(-1)^\ell$. Chang and Li give that mode-level form directly, together with the reality condition it implies [44]. The elliptic interpretation is Schrödinger’s [45]; Gibbons applied it to black holes [46]; Folacci and Sánchez computed the elliptic scalar field theory [47]; Parikh, Savonije and Verlinde gave dS/ $\mathbb{Z}_2$ its modern form [48]; ’t Hooft made the antipodal identification the centre of a black-hole S-matrix proposal [49–50].

S11.3 The codimension classification. Codimension one, timelike: in a de Sitter-sliced warp respecting the null energy condition, Karch and Randall’s inequality forces the crease tension negative [51], and in the three-brane realisation of Kogan et al. [52] the crease carries the model’s ghost. These obstructions apply to the stated warped and three-brane constructions; higher-codimension de Sitter branes have separate no-go results [53]. Codimension one, spacelike: a time mirror has no tension. Codimension two: the fixed set is the bifurcation sphere.

S11.4 The $2T_U$ crease state. The state respecting a codimension-two identification is the even Euclidean scalar on the π -angle $\mathbb{Z}_2$ cone, whose Green function is π -periodic, giving

$\Delta\langle\varphi^2\rangle = 1/(16\pi^2 r^2)$ and $\Delta\rho = 1/(32\pi^2 r^4)$; both coefficients are Moretti's [54]. Read Lorentzianly, the crease state is the Rindler thermal state at twice the Unruh temperature, so it is non-integrable at the horizon: singular exactly where it is a fold. Two independent checks: the conical formula $\langle\varphi^2\rangle = (N^2-1)/(48\pi^2 r^2)$ [55] at $N = 2$, and the Rindler thermal density $\rho = [(T/T_U)^4-1]/(480\pi^2 d^4)$ at $T = 2T_U$. Withdrawn in the same place: reading the deficit angle π as a Lorentzian positive-tension defect; a Euclidean deficit is an $SO(2)$ rotation while the real Lorentzian action lies in a disconnected component of the Lorentz group.

S11.5 Why four dimensions. Fold the bulk in half and the graviton sits exactly on the unitarity boundary, and it does so in four dimensions and no other.

For the Z_2 -odd transverse-traceless graviton of a dS_d -sliced AdS_{d+1} bulk, $(\sinh^d u \psi')' + \epsilon \sinh^{d-2} u \psi = 0$ with $\psi(L) = 0$ at the crease and $\psi'(w_0) = 0$ at the brane, the Higuchi/partially-massless value $\epsilon_{PM} = d - 2$ [56] is the lowest eigenvalue on the midpoint line $w_0 = 2L$ **only at $d = 4$** .

Two things have to line up, and only $d = 4$ makes both happen. The antiderivative $(d - 2) \int du / \sinh^{d-2} u$ must be degree one in $\coth u$, which holds at $d = 4$ alone; and the boundary term $1/(\sinh^{d-3} L \cosh L)$ must be a $\sinh$ of the doubled position, which it is $(2/\sinh 2L)$ only at $d = 4$. The condition then collapses to $2 \sinh X \cosh Y = 2 \cosh X \sinh Y$. Writing each side as $\sinh(X+Y) \pm \sinh(X-Y)$ leaves $2 \sinh(X-Y) = 0$, so $\sinh(w_0 - 2L) = 0$ and the crease sits at the midpoint.

Numerically ϵ_1 on the midpoint line at $L = 0.5, 1$ and 2 is 2.00000 three times at $d = 4$, against $2.998/3.631/6.977$ at $d = 3$, $1.291/1.033/0.480$ at $d = 5$ and $0.806/0.502/0.102$ at $d = 6$. solves the eigenvalue problem by shooting and reproduces all twelve; the $d = 4$ row comes out on the partially-massless value to a relative error of 3×10^{-10} at every L .

The model in which this happens has a ghost at that point, so it is a fact about a sick model and no observability claim is attached.

S11.6 The RP^4 one-loop invariant. Conformal Laplacian $L = -\nabla^2 + 2$ at unit radius; eigenvalues $\lambda_{\square} = (\square+1)(\square+2)$ with degeneracy $(\square+1)(\square+2)(2\square+3)/6$; the antipodal map acts as $(-1)^\ell$. $F = -\frac{1}{2}\zeta'(0) - \zeta(0)\ln \mu R$; the two parities' zeta functions add to the sphere's term by term, so $\Delta F_+ + \Delta F_- = 0$, and $\zeta(0) = -\chi/180$ in all three cases. Closed forms: $\zeta'_-\{S^4\}(0) = 1/144 + \zeta'(-1)/3 + 2\zeta'(-3)/3$ and $\zeta'_+(0) - \zeta'_-(0) = 7\zeta(3)/(4\pi^2)$. The reference value is $F_- \{S^4\} = -1/288 - \zeta'(-1)/6 - \zeta'(-3)/3 + (1/90)\ln \mu R$, its log coefficient checked against [57]. Why no counterterm absorbs it: every counterterm in the four-dimensional gravitational effective action is the integral of a local curvature scalar, and RP^4 is locally isometric to S^4 with exactly half its volume, so each contributes exactly half its S^4 value; Gauss-Bonnet included, since χ goes $2 \rightarrow 1$; and all of that sits inside $\frac{1}{2}F_- \{S^4\}$ and cancels in the difference. Seven numerical routes check these fixed spectral identities, with S^2 and S^3 controls. Their high-precision agreement concerns the calculation, not the accuracy of a physical prediction.

S11.7 The discrete-curvature reproduction. The 2024 background is $\dot{a}^2 = (\lambda/3)(a^4 - 2\tilde{\kappa}a^2 + 1)$, with $\tilde{\kappa} = 3\kappa/(2\lambda)$ and matching $G(\tilde{\kappa}) = \sqrt{2\tilde{\kappa}/3} I(\tilde{\kappa}) = N\pi/2$ or $\pi/(2N)$; the closed form is $I(\tilde{\kappa}) = \pi/[2 \operatorname{agm}(1, \sqrt{(1-\tilde{\kappa})/2})]$, with $I(0) = \Gamma(1/4)^2/(4\sqrt{\pi}) = 1.854074677301$ and $I(-1) = \pi/2$. Conversion $\Omega_K = -2\tilde{\kappa}\sqrt{\Omega_\Lambda\Omega_r}$; the difference between the two 2024 arXiv versions is entirely in that coefficient. The 2025 background has four components, $\tilde{\kappa}_{25} = (2/3)\tilde{\kappa}_{24}$ and matter $\rho_m \propto |a|^{-3}$, so the background stays symmetric across the bang, conversion $\Omega_K = -3\tilde{\kappa}_{25}\sqrt{\Omega_\Lambda\Omega_r}$; the matching tightens from $n\pi/2$ to $n\pi$, and the two calibrations differ by $2/\sqrt{3}$, decomposing as $\times 2$ from symmetric-only modes and $\times 1/\sqrt{3}$ because the 2024 phase is acoustic and the 2025 phase free-streaming. The matter-free limit of the 2025 integral reproduces the 2024 closed form to 4.4×10^{-16} .

S11.8 Three items deferred. (i) Barbour's own proposed general-relativistic complexity, the Yamabe invariant [20], does **not** detect the bang: it depends only on the conformal 3-geometry, which is frozen there; over the super-horizon window the comoving curvature

perturbation varies by 2.2×10^{-3} while $\square$ changes by factors of 10^3 - 10^6 . (ii) The linear ceiling: exactly, in the radiation era, $\mathcal{P}_S = 3\sqrt{2}\Phi_i|xj_1(x)|$, maximum $4.5104\Phi_i$ at $x = 2.7437$, so at the measured amplitude $\Phi_i \simeq 2 \times 10^{-5}$ the complexity is capped at about 10^{-4} ; Barbour's unbounded growth is nonlinear and invisible here. (iii) The equation-of-state ladder: with $\rho_w \propto a^{-(2N+2)/N}$, the CPT-allowed rungs are $N = 3$ ($w = -1/9$, $\rho \propto a^{-8/3}$), $N = 5$ ($w = -1/5$, $\rho \propto a^{-12/5}$) and $N = -1$ (a cosmological constant); every rung is shallower than a^{-4} .

S11.9 The Bel-Robinson complexity. With E and B the electric and magnetic parts of the Weyl tensor and observer $u = a^{-1}\partial_\eta$,

$$\mathcal{P} = \left[\frac{8\langle E_{ab}E^{ab} + B_{ab}B^{ab} \rangle}{(R_{ab}R^{ab})_0} \right]^{1/2},$$

normalised by $R_{ab}R^{ab}$ because the Ricci scalar is the exact constant 4λ for radiation plus Λ . The CPT condition on the tetrad [58] says, in conformal-Newtonian gauge, exactly that the Bardeen potential is **even**. So regularity at the bang and evenness are the same condition. The background is conformally flat, so the first-order Weyl tensor is gauge-invariant by Stewart and Walker [59]. The regular modes give $\mathcal{P}_S = (\sqrt{2}/3)k^2|\Phi_i|\eta^2$ and $\mathcal{P}_T = \sqrt{8/27}k^2|h_i|\eta^2$, exactly even on both sheets and vanishing at the bang from either side. The irregular Bardeen solution goes as η^{-3} and the irregular tensor mode as η^{-1} , so both give $\mathcal{P} \propto |\eta|^{-1}$.

Two of Barbour's legs transfer. $E_{\text{tot}} = 0 \leftrightarrow \kappa = 0$ is an identity, since κ is minus the total energy per unit mass. $L_{\text{tot}} = 0$ corresponds to vanishing primordial vorticity and is derived rather than assumed: the gauge-invariant vector perturbation obeys $\sigma'_g + 2\mathcal{H}\sigma_g = 0$, so $\sigma_g \propto \tau_-^2$ is even, while σ_g is also the derivative of an even tensor mode and hence odd. Both cannot hold unless the coefficient vanishes.

The leg that fails is $D = 0$: $d(a^2)/dt \rightarrow 2\sqrt{r/3}$ at the bang, a non-zero constant.

S11.10 The CP-wall threshold and its e-fold cost. Supplement §S3's 29.5 MeV and ΔN , derived. A wall of energy per unit area $\sigma = \Lambda_{\text{CP}}^3$ crossing our Hubble volume contributes $\rho_{\text{wall}} \sim \sigma/R_H$, and is harmless only if that is far below the critical density:

$$\sigma = \Lambda_{\text{CP}}^3, \quad \rho_{\text{wall}} \sim \frac{\sigma}{R_H} \ll \rho_c \iff \Lambda_{\text{CP}} < (\rho_c R_H)^{1/3}.$$

The inputs are $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $c = 2.99792458 \times 10^8 \text{ m s}^{-1}$, $G = 6.674 \times 10^{-11} \text{ SI}$, $\rho_c = 3H_0^2/(8\pi G)$ and $R_H = c/H_0$, with $1 \text{ Mpc} = 3.0857 \times 10^{22} \text{ m}$, $\hbar c = 1.9733 \times 10^{-16} \text{ GeV m}$ and $1 \text{ kg} = 5.6096 \times 10^{26} \text{ GeV}$. Thus

$$\begin{aligned} \rho_c &= 3.678 \times 10^{-47} \text{ GeV}^4, \\ R_H &= 6.955 \times 10^{41} \text{ GeV}^{-1}, \\ (\rho_c R_H)^{1/3} &= 29.5 \text{ MeV}. \end{aligned}$$

Above that threshold the wall must sit at

$$\frac{d}{R_H} = \frac{\sigma}{\rho_c R_H} = \left(\frac{\Lambda_{\text{CP}}}{29.5 \text{ MeV}} \right)^3, \quad \Delta N = 3 \ln \left(\frac{\Lambda_{\text{CP}}}{29.5 \text{ MeV}} \right).$$

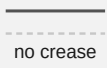
The threshold is the Zel'dovich-Kobzarev-Okun bound [5], recovered here from the wall required by §S3's imposed odd CP-field condition, not from a Kibble network, and the e-fold cost is the arithmetic of pushing a wall of that tension $(\Lambda_{\text{CP}}/29.5 \text{ MeV})^3$ Hubble radii outside

the horizon: $d/R_H = 3.91 \times 10^4$ at $\Lambda_{\text{CP}} = 1$ GeV, 3.91×10^{13} at 10^3 GeV and 3.91×10^{28} at 10^8 GeV, hence $\Delta N = 10.6$, 31.3 and 65.8 beyond the horizon-problem minimum, so that $N_{\text{total}} = 125.8$ at 10^8 GeV against $N_{\text{min}} = 60$. Every step is a statement about inflation and none of it is a statement about CP physics; the caveats stated in Supplement §S3 are the CT-odd assumption, the generic wall position, the neglected repulsive gravity of the wall, and that the Λ_{CP} values are illustrative, not predicted.

S12. Figures for supplementary mechanisms

Figure 1. Where a fold can sit, and what it costs

Shaded rows are consistent. Hatching marks a failed smooth one-copy fold.

fixed set	identification	price	verdict
 no crease	elliptic quotient dS/A free action: $A: x \rightarrow -x$	no crease at all; local correlators are the cover's restriction	invisible locally at tree level
 codimension 1	bang mirror $t \rightarrow -t$ fixes the spatial slice	mirror boundary condition; no tension; positivity needs a restricted atlas	best-studied fold
 codimension 2	modular fold dS/J (TP₁) fixes the bifurcation 2-sphere	Rindler at $2T^U$; $\Delta\rho = 1/(32\pi^2 d^4)$, divergent at the crease	dead as a smooth one-copy fold
 codimension 1	brane at fixed y +++ three-brane model	every dS crease: negative tension brane (Karch–Randall eq. 13, NEC); it carries a ghost	not a healthy fold
 two copies	glue at the bifurcation sphere second sheet is the J-image	eternal black hole in the Hartle–Hawking state	textbook: thermofield dynamics

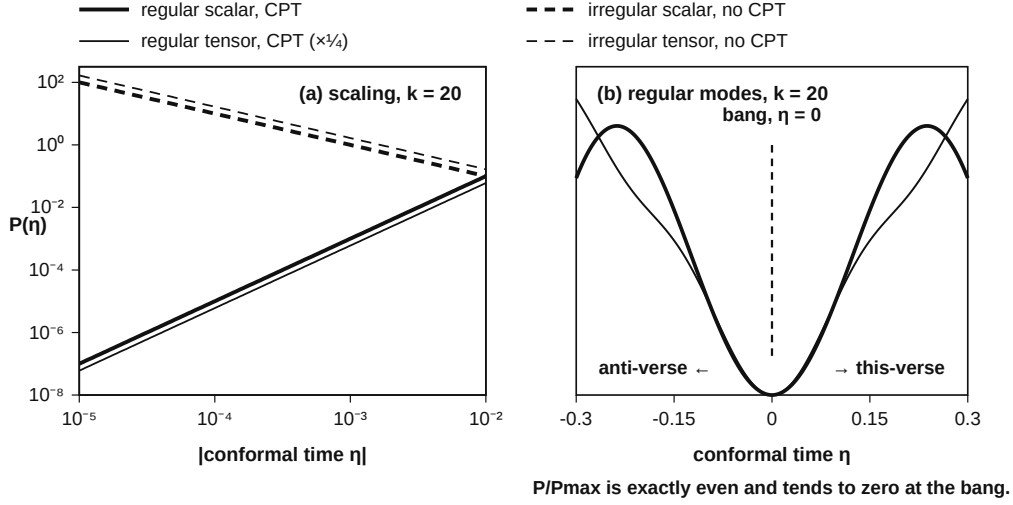
In the two-copy row, the cosmological-horizon analogue is the antipodal static patch.

Figure 1: the crease dictionary. The classification of §S11.1 drawn as a table: fold position, involution, fixed set, price, verdict.

Figure 2. The bang as a complexity minimum

Positive Bel–Robinson complexity $\mathcal{P}$. Solid: CPT-regular; dashed: irregular, no CPT.

Regular CPT modes: $\mathcal{P} \propto \eta^2$. Irregular no-CPT modes: $\mathcal{P} \propto |\eta|^{-1}$.



Regular modes vanish from both sides. Panel (a) scales the tensor by $\times 1/4$; panel (b) normalises to its scalar maximum.
Linear order only: the metric is degenerate at $\eta = 0$, so $\mathcal{P}(0) = 0$ is a two-sided limit, not a value at the bang.

Figure 2: the bang as a complexity minimum. The Bel–Robinson complexity $\mathcal{P}$ of §S4 against conformal time for the CPT-regular scalar and tensor modes ($\propto \eta^2$, vanishing at the bang from both sides) and for the irregular modes ($\propto |\eta|^{-1}$), shown for the representative case $k = 20$. The fitted exponents at $k = 5, 20, 80$ are reported in §S4 and §S11.9.

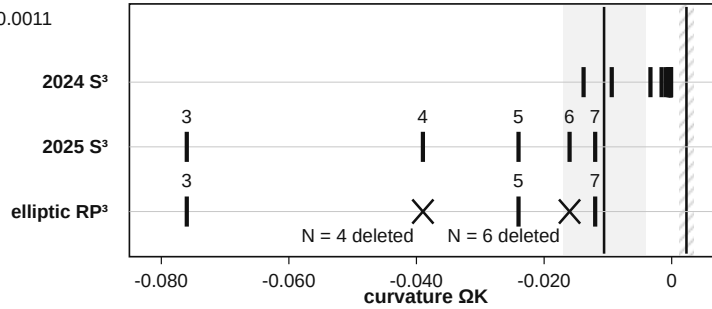
Figure 3. Discrete curvature against 2026 data

All listed allowed values are negative. Filled ticks are retained by RP³; crossed ticks are deleted.

data bands

Planck: -0.0106 ± 0.0065

DESI: $+0.0023 \pm 0.0011$



2025 set: $N = 3, 4, 5, 6, 7$ gives $\Omega_K = -0.076, -0.039, -0.024, -0.016, -0.012$.

Nearest allowed value: $N = 7$, $\Omega_K = -0.012$, 13.0σ from DESI DR2 + CMB, but 0.2σ from Planck alone.

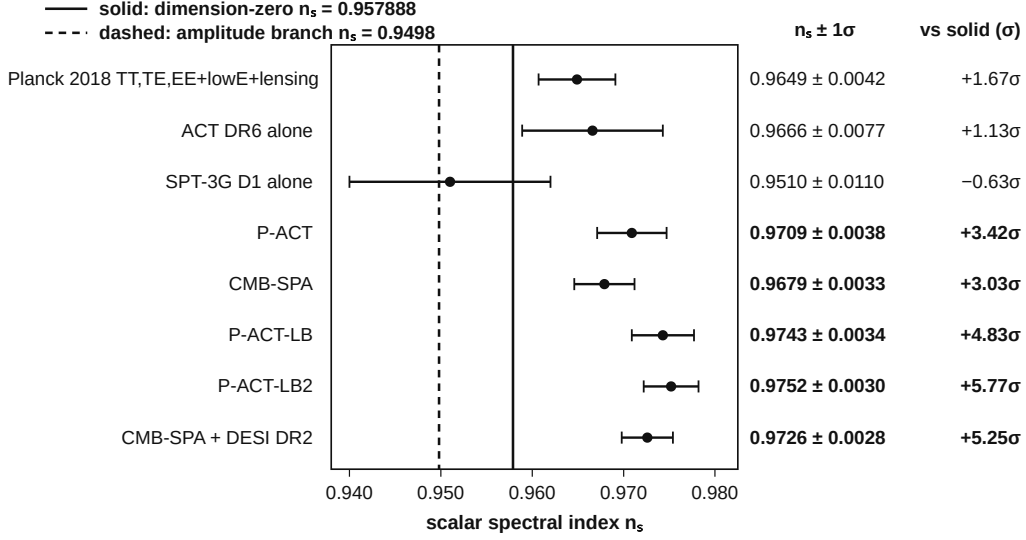
The 2025 χ^2 minimum $N = 4$ is 37.5σ from DESI DR2 + CMB and is exactly the value the elliptic reading forbids.

No compact reading admits positive Ω_K . Exact flatness is 2.09σ from DESI DR2 + CMB.

Figure 3: the quantised curvature against 2026 data. Deng and Handley’s allowed Ω_K set for $N = 3 \dots 7$ [24], with the members the elliptic reading deletes marked, against Planck 2018 alone [26] and DESI DR2 BAO + CMB [25]. Tensions as tabulated in §S5.

Figure 4. Primordial tilt against 2025–26 CMB data

All estimates are base- Λ CDM. Circles show the central estimate; bars show $\pm 1\sigma$.



Prediction: $n_s = 1 - 7\alpha_s(M_p)/\pi = 0.957888$. Amplitude at $N_0 = 36$ with Standard-Model couplings demands $n_s = 0.9498$.

The two values are not on the same one-parameter locus. This is posterior tension with the mechanism, not a formal exclusion.

The CPT-symmetric structure fixes no tilt. Freeing N_{eff} removes the tension, but needs $\Delta N_{\text{eff}} \approx -0.35$ to -0.82 .

Figure 4: the primordial tilt against 2026 data. $n_s = 0.957888$ from Turok and Boyle’s dimension-zero mechanism [30] against the eight datasets of §S6 [26,32–34], each shown with its displacement in σ .

S13. A weak Yukawa decay benchmark

The example in main §2.4 retains a fixed rank-two tree-level seesaw from two heavier, non-degenerate sterile neutrinos and adds only the small column y_α of N_1 . Its free lifetime and flavour direction are not selected by geometry. All formulas below are at leading seesaw and decay order, with $H^0 = (v + h + iG^0)/\sqrt{2}$, $v = 246.22$ GeV and $q = y^\dagger y$.

For $r_b = m_b^2/M_1^2$, define $f_V(r) = (1 - r)^2(1 + 2r)$ and $f_h(r) = (1 - r)^2$. Averaging over the N_1 spin and summing over both light-neutrino helicities in the neutral channels and both charges in the $W\ell$ channel gives [60]

$$\Gamma_0 = \frac{qM_1}{32\pi} \{2f_V(r_W) + f_V(r_Z) + f_h(r_h)\} \simeq \frac{qM_1}{8\pi}.$$

These are Born hard-process fractions, not exclusive no-radiation fractions. Inclusive widths receive perturbative corrections, and electroweak showering must be included in an observed spectral calculation [61]. We use $\tau = \hbar/\Gamma_0$ only as a freely chosen leading-order lifetime parameter.

If u_0 is a unit null vector of the rank-two mass matrix m_0 , the added complex symmetric matrix is $\delta m = -v^2 y y^T / (2M_1)$, with spectral norm $v^2 q / (2M_1)$. The Takagi masses are singular values, so

$$s_{\min}(m_0 + \delta m) \leq \|(m_0 + \delta m)u_0\| \leq \|\delta m\|_2 = \frac{4\pi v^2 \hbar}{M_1^2 \tau}.$$

Choosing $y = \sqrt{q} u_0^*$ attains this bound while leaving the two nonzero masses unchanged at this order. Choosing $u_0^T y = 0$ leaves a zero at this order. Thus the bound is not a flavour-independent nonzero prediction. Baseline radiative masses, running and matching corrections are outside this tree-level bound.

For $M_1 = 4.848 \times 10^8$ GeV and the illustrative $\tau = 10^{28}$ s, $q \simeq 3.4 \times 10^{-60}$. In standard radiation domination with $g_* = g_{*s} = 106.75$, conserved entropy and no other portal, $H(T = M_1) \simeq 0.33$ GeV and $\Gamma_0/H(T = M_1) \simeq 2.0 \times 10^{-52}$. A perturbative relativistic scattering rate CqT obeys $\Gamma_{\text{sc}}/H \simeq 5.0 \times 10^{-51} C$ at $T = M_1$ and falls as $1/T$ at higher temperatures. The integrated probability $\int_{T \geq M_1} (\Gamma_{\text{sc}}/H) d \ln T$ is at most this value for bounded C ; no ultraviolet power enhancement occurs in this renormalizable interaction.

An inverse-decay abundance check is stronger than nonthermalisation alone. In the Maxwell-Boltzmann approximation with $x = M_1/T$ and $g_N = 2$,

$$\frac{dY}{dx} = \frac{45g_N}{4\pi^4 g_{*s}} \frac{\Gamma_0}{H(T = M_1)} x^3 K_1(x), \quad Y = \frac{135g_N}{8\pi^3 g_{*s}} \frac{\Gamma_0}{H(T = M_1)} \simeq 2.0 \times 10^{-54},$$

using $\int_0^\infty x^3 K_1(x) dx = 3\pi/2$. The adopted present density and entropy inputs give $Y_{\text{DM}} = \rho_{\text{DM},0}/(s_0 M_1) \simeq 9.0 \times 10^{-19}$. Thermal-mass and statistical corrections cannot compensate for this coupling suppression in the stated perturbative bath. For a scattering reaction density $C_{\text{prod}} q T^4$, the relativistic yield is of order $10^{-52} C_{\text{prod}}$ or smaller. The depletion over 13.8 Gyr is $1 - e^{-t_0/\tau} \simeq 4.4 \times 10^{-11}$.

These estimates exhibit an internally consistent, effectively unchanged abundance at the quoted accuracy. They do not construct the interacting BFT initial state or constrain unspecified additional portals. Setting $y = 0$ restores the imposed species parity, explaining perturbative stability of a chosen tiny column without selecting its magnitude. If that parity is instead required to be an exact gauged symmetry, the added term needs a separate symmetry-breaking completion. The example assumes a weakly broken matter rule and makes no observational lifetime-bound or KM3NeT event-rate claim.

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The supplement compiles as its own document, so it carries its own list and its own numbering. Neither follows the main paper's.

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